Preschool Learning with Artificial Intelligence (PLAI): Exploring Children´s Interaction with AI Toys and Robots*

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Abstract

The project PLAI work combines qualitative and quantitative methods to analyse children's interaction with intelligence toys and robots from an educational and psychological point of view. Current trends in digital play research focus on the effect of learning tools (i.e. video games and intelligence robots) on children's development and/or learning skills. This paper reports on the work in progress of exploring children's interaction with artificially intelligence toys and robots and their potential contribution to children´s learning journey. Furthermore, this study examines how these technologies facilitate the interactive learning process in educational settings. In the first stage, the project aims to build a conceptual framework for children's digital play and digital games-based learning experiences with these devices. In the second stage, the study of the AI features of toys and robots that potentially improve learning experiences is conducted through a content analysis – a specific instrument is developed to examine AI features. The final stage will involve performing a field study on child-robot interaction in a natural setting – a technique of gesture and movement analysis is employed for the recorded video of child-robot interaction.

CCS Concepts

• Human-centered computing; **• Human computer interaction (HCI)**;; **• Interaction paradigms**; **• Interaction devices.**;

Keywords

Artificial intelligence (AI), AI toys and robots, developmental domains, digital play, and child-robot interaction

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1 Introduction

The advancement of digital technology and the emergence of Artificial Intelligence (AI) have sparked significant discussions in various fields, including health, education, economics, politics, and social domains [1]. Children's lives with digital technology are no exception. AI technology has been immersed for instance in their play time. The definition of AI has been growing depending on the context and advancement of technology embedded in certain fields. AI is defined as a branch of computer science that combines algorithms, machine learning, and natural language processing [2]. Some AI features can be found in digital educational games or robots where it can store children's learning performance and process it to suggest personalized learning based on their learning analytics and preference [3].

Digital play refers to various activities where toys and other objects are mediated – where interaction between materials and immaterial practices are integrated in both physical and virtual spheres [4, 5]. This includes play that is mediated by technology as well as more traditional or modern forms of play that do not rely on technological tools [6]. Previous studies in digital play related to the use of internet of toys, intelligence toys and robots, and screenplay (video games with tablet, desktop, and phone) in different settings focus on children´s ecological aspects (i.e. home and school) [7, 8], play affordance with digital tools in outdoor context [9], literacy skills [10, 11], and math and reading skills [12]

Our previous systematic review has shown patterns of the commonly used educational games and intelligence toys or robots for children to develop specific skills (i.e. literacy, math, and reading) [13]. However, there is a lack of studies exploring the use of advanced technologies (i.e. educational games and intelligence toys) for physical-motor and socio-emotional skills taking place in natural settings, such as school and home. Therefore, this current study focuses on exploring the use of AI toys and robots, in natural settings through 3 stages: first, systematic review on the conceptual framework of digital play and digital games-based learning, second, content analysis of AI features on toys and robots for children under six years old and third, children's interaction with intelligence toys and robots in a school setting.

1.1 AI Toys and Robots

The AI toys and robots for children in the market are designed using certain levels of AI systems (machine learning, deep learning, natural language processing, and computer vision) that enable advanced tasks like face and emotion recognition, voice commands into robot action, personalized learning, and multilingual language features [14]. Besides the AI system, the physical characteristics are identified based on their components (i.e. camera, speaker, accessories to enhance play experience, and guidance book), colours and textures, dimension and weight, and movement and gestures [15].

There are five aspects of AI system that have been considered during the creation of instrument for the content analysis, they are: emotional-AI (a system that can detect the emotional expression of humans) [16], learning-AI (a system that stores and analyse children's data, called performance analytic [17] in the intent of personalizing the learning process) [18], command/sense-AI (a system that can operate and navigate gesture and voice) [19], network-AI (a system that enables user to interact and connect with others and/or with the toys [20], and data use (the collection of personal data and its capacity to interpret, process, and share it) [21].

1.2 Objectives

The study aims to explore children´s digital play experience using intelligence toys and robots that potentially contribute to their development through educational activities.

Specifically, this current study state three research questions:

- What are AI features embedded in children's toys and robots?
- Which are AI features in children's toys and robots that potentially promote educational activities, children's development and learning skills?
- How do children engage with AI toys and robots in a natural setting?

2 STUDY DESIGN

This research consists of three stages, the project PLAI started in October 2023.

2.1 Systematic Review

The conceptual framework of both digital play (DP) and digital games-based learning (DGBL) in early childhood education is not well defined and is unclear in terms of specific materials and tools (i.e. types of digital technology devices and objects children use), space and setting (home, school, and play space), their effects on children's developmental domains and learning skills, as well as roles of teachers and/or caregivers in these two practices (DP and DGBL). Therefore, a systematic review aims to build a solid understanding of the abovementioned concepts. It was done from November 2023 to January 2024 following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) [22] approach and complement with aggregative synthesis logic, as this logic aims to investigate studies that present educational interventions and propose the effect and impact on learners [23]. Our systematic review shows that DP appears in natural settings, such as at home, school and play space, which include tools, devices, and systems that can create, generate, store, and/or process data [24]. On the other hand, DGBL practice mostly happens in classroom settings and laboratories where specific intelligence toys and robots are integrated to see its effect on children's learning outcomes. Additionally, there has been a lack of studies exploring children´s interaction with intelligence toys and robots in relation to physical-motor developments in natural settings. The systematic review finished in January, and the results are one published as a book chapter, and another is still under review in the Education 3 to 13 journal.

2.2 Content Analysis of AI Toys and Robots

A thematic content analysis has been used in qualitative research to understand specific subjects with thick descriptive data [25]. This method is centred around traditional media studies, such as magazines and newspapers. However, it has evolved to more interactive media, such as interactive gameplay [26, 27]. In this stage, structured observation of content analysis is used to investigate the toys and robot's physical characteristics and AI features. The researcher developed an observation instrument covering three aspects: physical characteristics (18 items), AI functionalities (17 items), and children´s developmental domains (16 items). The instrument is validated in two stages: the first draft was validated by experts in children's development and AI technicalities, and the second stage was validated by four experts in children's health and well-being, AI and robotics, teaching pedagogy, and research methodology. Two-stage pilot coding was conducted to measure the instrument's reliability prior to the main inspection (by the principal researcher). In total, 20 intelligence toys and robots are being inspected physically (in the lab) and virtually (based on information available online). The content analysis was conducted from September to December 2024 and is now in the process of writing the report for publication.

2.3 Field work (next stage)

After obtaining results from the content analysis of intelligence toys and robots, the final stage is to explore and investigate how children interact with them in a classroom setting. Previous studies mention that the interaction of children and AI is a unique area to explore [28], considering there is a lack of studies focusing on children under six years old [29]. Hence, this stage is crucial and worth taking. As a part of the research design, this stage uses thematic observation and gesture and movement interaction analysis. Observing behaviours is the approach that helps the researcher establish meaning from the phenomenon from the participant's point of view [30], in this case, from the point of view of children with intelligence toys and robots. The toy robots are used to incorporate learning activities; hereafter, they fall under the condition of Digital Games-based learning approach practice. An observation guideline and checklist are developed based on the previous results of analysing toys and robots and their aspects related to AI features and children's developmental stages and skills. Apart from direct observation, children's activities with the toys and robots will be recorded to analyse further the complex gesture engagement and interaction between children with the intelligence toys and robots [31]. Detailed techniques in automatic gesture and movement analysis are a work in progress between the main researcher and a collaborative partner expert in children's physical-motor development. The fieldwork preparation started in March 2025 for document preparation, such as requesting ethical approval from the Universitat Oberta de Catalunya ethics committee and sending requests for initial meetings with school directors. The data collection in school is expected to happen in the fall semester, September 2025.

3 Publication

Book chapter

A. Utami and L. Crescenzi-Lanna, "Una revisión sistemática del juego digital y el aprendizaje basado en juegos en la Educación Infantil," in *Conectados y conscientes: Pantallas, educación y salud*, Ó. Flores Alarcia and T. Hernández Jover, Eds. Editorial, pp. 59–73, 2024. Available: https://dialnet.unirioja.es/servlet/articulo?codigo=9680896

Article journal (under review)

A. Utami and L. Crescenzi-Lanna, "Research Design and Implementation of Digital Play and Digital Games-based Learning in Early Development (0 to 6 years): A Systematic Review", 2025. *Education 3 to 13.*

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